Griffith UNIVERSITY

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Balance Laws in Continuum Mechanics

Mass, Linear and Angular Momentum, Energy, and Entropy

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1 Overview

The balance laws considered are:

- Conservation of mass
- Conservation of linear momentum
- Conservation of angular momentum
- 1st law of thermodynamics (balance of energy)
- 2nd law of thermodynamics (balance of entropy) — gives a local inequality, used for thermodynamically consistent constitutive material models

Definition 1.1 — Global and Local Form. A balance law has a **global (integral)** form and a **local (differential)** form. The global form is stated as an integral over a cut-out part of the body, either a **referential cut** $B_p \subset \mathcal{B}$ or a **spatial cut** $S_p \subset \varphi_t(\mathcal{B})$.

Definition 1.2 — Referential and Spatial Cuts. Let $\varphi : \mathcal{B} \to \varphi_t(\mathcal{B})$ be the deformation map taking a material point $\boldsymbol{X}$ in the reference configuration $\mathcal{B}$ to its position $\boldsymbol{x} = \varphi(\boldsymbol{X}, t)$ in the current configuration $\varphi_t(\mathcal{B})$ at time t. B_p is the referential (Lagrangian) cut-out part; $S_p = \varphi_t(B_p)$ is its spatial (Eulerian) image, with outward unit normal $\hat{\boldsymbol{n}}$.

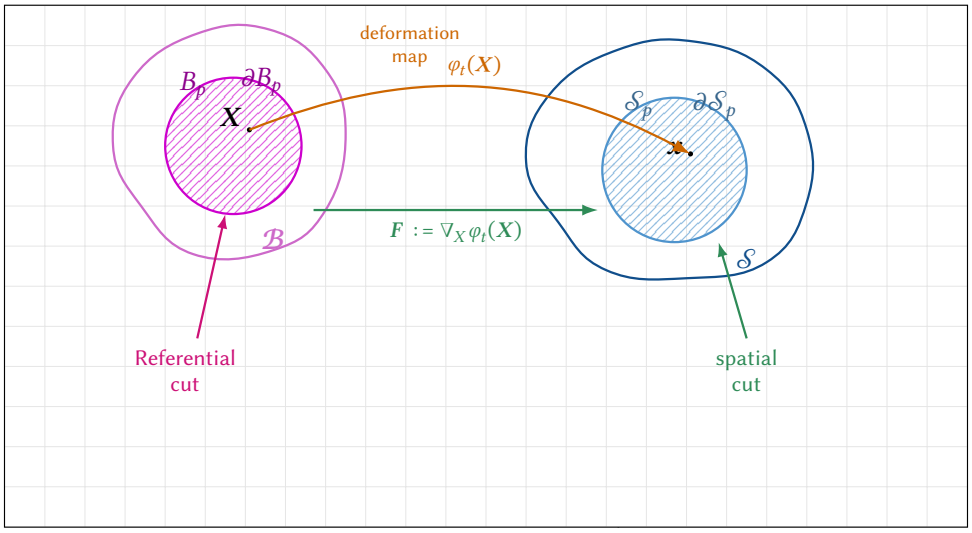


Figure 1: The referential cut-out part B_p and its spatial image, related by the deformation map $\varphi_t(\boldsymbol{X})$.

1.1 Transforming a Global Balance Law to Local Form

For the integral form of a balance law stated for the cut-out B_p or S_p:

- Apply a Cauchy-type theorem to replace the flux terms in the surface integrals.
- Apply Gauss' theorem to transform surface integrals to volume integrals.
- Apply the localisation theorem.

A **closed system** is one with no mass through the surface and no mass produced/destroyed within the domain.

2 The Global-to-Local Tools

Theorem 2.1 — Cauchy-Type Theorems. On the boundary of the spatial cut S_p, the surface traction $\boldsymbol{t}$ and the outward heat flux depend on the surface orientation only through the outward unit normal $\hat{\boldsymbol{n}}$, and this dependence is linear:

$$\boldsymbol{t}(\hat{\boldsymbol{n}}) = \boldsymbol{\sigma} \cdot \hat{\boldsymbol{n}}, \qquad q(\hat{\boldsymbol{n}}) = \boldsymbol{q} \cdot \hat{\boldsymbol{n}}, \tag{1}$$

where $\boldsymbol{\sigma}$ is the spatial stress tensor and $\boldsymbol{q}$ is the spatial heat flux vector.

Theorem 2.2 — Divergence (Gauss') Theorem. For a vector field $\boldsymbol{w}$ (or tensor field $\boldsymbol{T}$) on the spatial cut S_p

with boundary ∂S_p and outward normal $\hat{\boldsymbol{n}}$,

$$\int_{\partial S_p} \boldsymbol{w}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}a = \int_{S_p} \nabla\cdot\boldsymbol{w}\,\mathrm{d}v, \qquad \int_{\partial S_p} \boldsymbol{T}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}a = \int_{S_p} \nabla\cdot\boldsymbol{T}\,\mathrm{d}v. \tag{2}$$

Theorem 2.3 — Localisation Theorem. If a continuous field f satisfies $\int_{S_p} f\,\mathrm{d}v = 0$ for every choice of cut-out S_p, then $f \equiv 0$ at every point.

2.1 The Closed-System Identity

Mass is conserved, so for a closed system no mass crosses ∂S_p: $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}m) = \dfrac{\mathrm{d}}{\mathrm{d}t}(\rho\,\mathrm{d}v) = 0$. Hence for any field ϕ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{S_p}\rho\,\phi\,\mathrm{d}v = \frac{\mathrm{d}}{\mathrm{d}t}\int_{S_p}\phi\,\mathrm{d}m = \int_{S_p}\dot{\phi}\,\mathrm{d}m = \int_{S_p}\rho\,\dot{\phi}\,\mathrm{d}v. \tag{3}$$

3 Conservation of Mass

Definition 3.1 — Mass of the Referential and Spatial Cuts. Let $\rho_0(\boldsymbol{X})$ be the mass density in the reference configuration and $\rho(\boldsymbol{x},t)$ the mass density in the current configuration. The mass of the referential cut B_p and of the spatial cut $S_p = \varphi_t(B_p)$ are

$$M_{B_p} = \int_{B_p}\rho_0(\boldsymbol{X})\,\mathrm{d}V, \qquad M_{S_p} = \int_{S_p}\rho(\boldsymbol{x},t)\,\mathrm{d}v. \tag{4}$$

Since $\mathrm{d}v = J\,\mathrm{d}V$, with $J = \det\boldsymbol{F}$, the equality $M_{B_p} = M_{S_p}$ reads $\int_{B_p}\rho_0\,\mathrm{d}V = \int_{B_p}\rho J\,\mathrm{d}V$ for an arbitrary B_p. Localisation (Theorem 2.3) gives

$$\rho_0(\boldsymbol{X}) = \rho(\boldsymbol{x},t)\,J(\boldsymbol{X},t). \tag{5}$$

Since B_p is a fixed piece of the reference body, ρ_0 carries no explicit time dependence, so $\dfrac{\mathrm{d}}{\mathrm{d}t}M_{B_p} = 0$. Mass conservation therefore also requires $\dfrac{\mathrm{d}}{\mathrm{d}t}\displaystyle\int_{B_p}\rho J\,\mathrm{d}V = 0$. Localisation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(\rho J) = \dot{\rho}J + \rho\dot{J} = 0. \tag{6}$$

Lemma 3.0.1 — Rate of the Jacobian. $\dot{J} = J\,\nabla\cdot\boldsymbol{v}$.

Proof Sketch. $\dot{J} = \dfrac{\mathrm{d}}{\mathrm{d}t}\det(\boldsymbol{F}) = J\,\mathrm{tr}(\dot{\boldsymbol{F}}\boldsymbol{F}^{-1})$. Since $\dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = \nabla\boldsymbol{v} = \boldsymbol{L}$, the spatial velocity gradient, and $\mathrm{tr}(\boldsymbol{L}) = \nabla\cdot\boldsymbol{v}$.

Substituting Lemma 3.0.1 and dividing by $J \neq 0$ yields the **continuity equation**:

$$\boxed{\dot{\rho} + \rho\,\nabla\cdot\boldsymbol{v} = 0}. \tag{7}$$

For an incompressible material, $\dot{\rho} = 0$, and the continuity equation reduces to $\nabla\cdot\boldsymbol{v} = 0$.

4 Conservation of Linear Momentum

Definition 4.1 — Linear Momentum Balance. The linear momentum of the spatial cut S_p is $\boldsymbol{L}_{S_p} = \int_{S_p}\rho\boldsymbol{v}\,\mathrm{d}v$. The rate of change of linear momentum equals the net force acting on S_p, i.e. the body force (acting at a distance) plus the surface force (acting in the vicinity):

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{S_p}\rho\boldsymbol{v}\,\mathrm{d}v = \int_{S_p}\rho\boldsymbol{b}\,\mathrm{d}v + \int_{\partial S_p}\boldsymbol{t}\,\mathrm{d}a, \tag{8}$$

where $\boldsymbol{b}$ is the body force per unit mass in the current configuration and $\boldsymbol{t}$ is the Cauchy surface stress traction

vector.

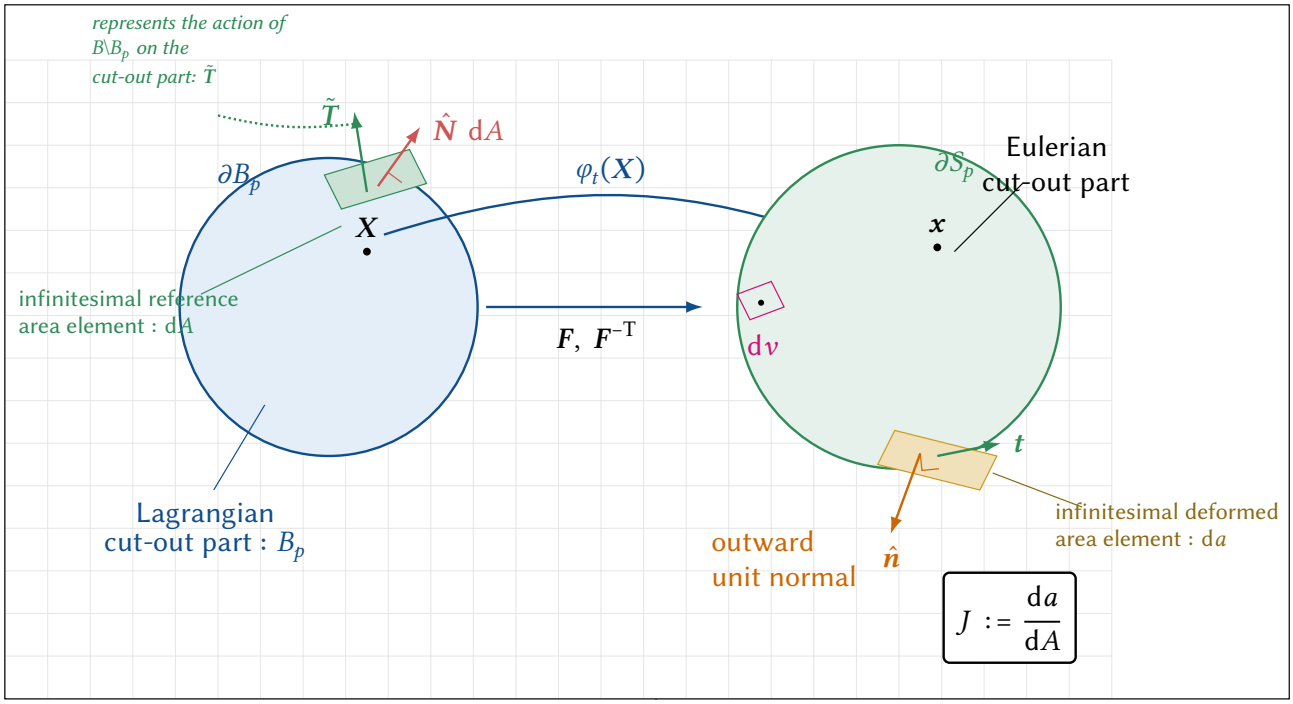


Figure 2: The Lagrangian cut-out part B_p and its Eulerian image S_p, with the infinitesimal reference and deformed area elements $\mathrm{d}A$, $\mathrm{d}a$, their normals $\hat{\mathbf{N}}$, $\hat{\boldsymbol{n}}$, the traction $\boldsymbol{t}$, and the area-ratio Jacobian $J = \mathrm{d}a/\,\mathrm{d}A$.

Since mass is conserved, $\dfrac{\mathrm{d}}{\mathrm{d}t}M_{S_p} = 0$, so the closed-system identity (3) applies with $\phi = \boldsymbol{v}$ (so $\dot{\boldsymbol{v}} = \boldsymbol{a}$). Using Cauchy's theorem (1) to write $\boldsymbol{t} = \boldsymbol{\sigma}\cdot\hat{\boldsymbol{n}}$, and the divergence theorem (2):

$$\int_{S_p}\rho\boldsymbol{a}\,\mathrm{d}v = \int_{S_p}\rho\boldsymbol{b}\,\mathrm{d}v + \int_{S_p}\nabla\cdot\boldsymbol{\sigma}\,\mathrm{d}v \quad\Longrightarrow\quad \int_{S_p}(\nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{b} - \rho\boldsymbol{a})\,\mathrm{d}v = 0. \tag{9}$$

The cut-out S_p is arbitrary, so the localisation theorem (Theorem 2.3) gives the local form of the balance of linear momentum:

$$\boxed{\nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{b} = \rho\boldsymbol{a}}. \tag{10}$$

5 Conservation of Angular Momentum

Definition 5.1 — Angular Momentum Balance. The angular momentum of the cut-out S_p is $\boldsymbol{H}_{S_p} = \int_{S_p}\boldsymbol{x}\times\rho\boldsymbol{v}\,\mathrm{d}v$. Its rate of change equals the resultant moment generated by the body and surface forces acting on S_p:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{S_p}\boldsymbol{x}\times\rho\boldsymbol{v}\,\mathrm{d}v = \int_{S_p}\boldsymbol{x}\times\rho\boldsymbol{b}\,\mathrm{d}v + \int_{\partial S_p}\boldsymbol{x}\times\boldsymbol{t}\,\mathrm{d}a. \tag{11}$$

The closed-system identity (3) applied to $\phi = \boldsymbol{x}\times\boldsymbol{v}$ gives $\dot{\phi} = \boldsymbol{v}\times\boldsymbol{v} + \boldsymbol{x}\times\boldsymbol{a} = \boldsymbol{x}\times\boldsymbol{a}$, so the left side of (11) becomes $\int_{S_p}\boldsymbol{x}\times\rho\boldsymbol{a}\,\mathrm{d}v$. Writing the surface term in index notation with $t_k = \sigma_{kl}n_l$ and applying the divergence theorem,

$$\int_{\partial S_p}\varepsilon_{ijk}x_j\sigma_{kl}n_l\,\mathrm{d}a = \int_{S_p}\varepsilon_{ijk}\frac{\partial(x_j\sigma_{kl})}{\partial x_l}\,\mathrm{d}v = \int_{S_p}\varepsilon_{ijk}\big(\sigma_{kj} + x_j\,(\nabla\cdot\boldsymbol{\sigma})_k\big)\,\mathrm{d}v. \tag{12}$$

Substituting back into (11) in index form,

$$\int_{S_p}\varepsilon_{ijk}x_j\rho a_k\,\mathrm{d}v = \int_{S_p}\varepsilon_{ijk}x_j\rho b_k\,\mathrm{d}v + \int_{S_p}\varepsilon_{ijk}\big(\sigma_{kj} + x_j(\nabla\cdot\boldsymbol{\sigma})_k\big)\,\mathrm{d}v, \tag{13}$$

and using Cauchy's equation of motion (10) pointwise, $\rho a_k = \rho b_k + (\nabla \cdot \boldsymbol{\sigma})_k$, the terms containing $\boldsymbol{x}$ cancel, leaving

$$\int_{S_p} \varepsilon_{ijk}\sigma_{kj}\,\mathrm{d}v = 0 \tag{14}$$

for an arbitrary S_p. Localisation (Theorem 2.3) forces $\varepsilon_{ijk}\sigma_{kj} = 0$ for every i, which is the statement $\sigma_{jk} = \sigma_{kj}$:

$$\boxed{\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}}. \tag{15}$$

6 Balance of Energy: The First Law of Thermodynamics

Definition 6.1 — Energy of the Spatial Cut. The total energy of the sub-body S_p is the sum of kinetic energy K and internal energy U,

$$\mathcal{E}_{S_p} = K + U, \qquad K = \int_{S_p} \tfrac{1}{2}\rho\, \boldsymbol{v}\cdot\boldsymbol{v}\,\mathrm{d}v, \qquad U = \int_{S_p} \rho e\,\mathrm{d}v, \tag{16}$$

with e the mass-specific internal energy density. The rate of change of the total energy equals the external mechanical power input $\mathcal{P}_{\mathrm{ext}}$ (rate at which the surface traction $\boldsymbol{t}$ and body force $\boldsymbol{b}$ perform work on S_p) plus the thermal power $\mathcal{Q}$ (rate of heat input — the outward flux through ∂S_p plus a distributed internal heat source of intensity r per unit mass):

$$\dot{\mathcal{E}}_{S_p} = \mathcal{P}_{\mathrm{ext}} + \mathcal{Q}, \qquad \mathcal{P}_{\mathrm{ext}} = \int_{S_p} \rho\boldsymbol{b}\cdot\boldsymbol{v}\,\mathrm{d}v + \int_{\partial S_p} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}a, \qquad \mathcal{Q} = \int_{S_p} \rho r\,\mathrm{d}v - \int_{\partial S_p} \boldsymbol{q}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}a. \tag{17}$$

Applying the divergence theorem to the traction term of $\mathcal{P}_{\mathrm{ext}}$,

$$\int_{\partial S_p} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}a = \int_{\partial S_p} (\boldsymbol{\sigma}\cdot\hat{\boldsymbol{n}})\cdot\boldsymbol{v}\,\mathrm{d}a = \int_{S_p} \left[(\nabla\cdot\boldsymbol{\sigma})\cdot\boldsymbol{v} + \boldsymbol{\sigma} : \boldsymbol{L}\right]\mathrm{d}v. \tag{18}$$

Using $\nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{b} = \rho\boldsymbol{a}$ and the closed-system identity (3),

$$\mathcal{P}_{\mathrm{ext}} = \int_{S_p} \rho\boldsymbol{a}\cdot\boldsymbol{v}\,\mathrm{d}v + \int_{S_p} \boldsymbol{\sigma} : \boldsymbol{L}\,\mathrm{d}v = \dot{K} + \int_{S_p} \boldsymbol{\sigma} : \boldsymbol{D}\,\mathrm{d}v, \tag{19}$$

where $\boldsymbol{D} = \frac{1}{2}(\boldsymbol{L} + \boldsymbol{L}^{\mathrm{T}})$ is the symmetric part of the velocity gradient (the stress power uses $\boldsymbol{D}$ rather than $\boldsymbol{L}$ because $\boldsymbol{\sigma}$ is symmetric, Equation (15), and contracts to zero against the skew part of $\boldsymbol{L}$).

6.1 Theorem of Expended Power

Theorem 6.1 — Theorem of Expended Power. Define the stress power $\mathcal{P}_{\mathrm{int}} = \int_{S_p} \boldsymbol{\sigma} : \boldsymbol{D}\,\mathrm{d}v$. Then

$$\mathcal{P}_{\mathrm{ext}} = \dot{K} + \mathcal{P}_{\mathrm{int}}. \tag{20}$$

The total mechanical power due to the externally applied forces equals the rate of change of kinetic energy plus the stress power of the cut-out S_p.

6.2 Localisation: The Local Energy Equation

Substituting $\mathcal{P}_{\mathrm{ext}} = \dot{K} + \mathcal{P}_{\mathrm{int}}$ into the first law (17) and $\dot{\mathcal{E}}_{S_p} = \dot{K} + \dot{U}$ cancels $\dot{K}$ from both sides:

$$\dot{U} = \mathcal{P}_{\mathrm{int}} + \mathcal{Q}, \qquad \text{i.e.} \qquad \frac{\mathrm{d}}{\mathrm{d}t}\int_{S_p} \rho e\,\mathrm{d}v = \int_{S_p} \boldsymbol{\sigma} : \boldsymbol{D}\,\mathrm{d}v + \int_{S_p} \rho r\,\mathrm{d}v - \int_{\partial S_p} \boldsymbol{q}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}a. \tag{21}$$

This is the decoupling of the mechanical and thermal power into their associated counterparts (internal energy and stress power/heat, respectively) — i.e. the conservation of energy, thermal and mechanical. Applying the closed-system identity to the left side and the divergence theorem to the heat-flux term, then localising over the arbitrary cut S_p, gives the local form of energy balance:

$$\boxed{\rho\dot{e} = \boldsymbol{\sigma} : \boldsymbol{D} - \nabla\cdot\boldsymbol{q} + \rho r}. \tag{22}$$

7 Balance of Entropy: The Second Law of Thermodynamics

The first law of thermodynamics indicates the balance of energy but does not place a restriction on the direction of the process. In heat transfer, the preferred direction of the spontaneous heat flow between two systems at distinct temperatures is from the warmer system to the colder one. Thus, there is a preferred direction, and this limitation is generalised and mathematically expressed as an inequality: the internal entropy production is always non-negative, and it is positive for an irreversible process. Entropy is a measure of disorder, work conjugate to temperature (like stress and strain).

Definition 7.1 — Entropy Balance. Let $\eta(\boldsymbol{x}, t)$ be the mass-specific entropy density and $\theta(\boldsymbol{x}, t) > 0$ the absolute temperature. The total entropy of the sub-body S_p is $H_{S_p} = \int_{S_p} \rho\eta \, \mathrm{d}v$. The second law states that the rate of entropy increase in S_p is bounded below by the entropy input rate $\dot{\Sigma}_{S_p}$, with an internal heat source r:

$$\dot{H}_{S_p} \geq \dot{\Sigma}_{S_p}, \qquad \dot{\Sigma}_{S_p} = \int_{S_p} \frac{\rho r}{\theta} \, \mathrm{d}v - \int_{\partial S_p} \frac{\boldsymbol{q} \cdot \hat{\boldsymbol{n}}}{\theta} \, \mathrm{d}a. \tag{23}$$

The total rate of entropy production within the sub-body S_p is $\Gamma = \dfrac{\mathrm{d}}{\mathrm{d}t} \displaystyle\int_{S_p} \rho\eta \, \mathrm{d}v - \dot{\Sigma}_{S_p} \geq 0$. Applying the closed-system identity to $\dot{H}_{S_p}$ and expanding the divergence term using $\nabla \cdot (\boldsymbol{q}/\theta) = \nabla \cdot \boldsymbol{q}/\theta - \boldsymbol{q} \cdot \nabla\, \theta/\theta^2$:

$$\int_{S_p} \rho\dot{\eta} \, \mathrm{d}v \geq \int_{S_p} \frac{\rho r}{\theta} \, \mathrm{d}v - \int_{S_p} \frac{\nabla \cdot \boldsymbol{q}}{\theta} \, \mathrm{d}v + \int_{S_p} \frac{\boldsymbol{q} \cdot \nabla\, \theta}{\theta^2} \, \mathrm{d}v. \tag{24}$$

Localising over the arbitrary cut S_p gives the local **Clausius–Duhem inequality**:

$$\boxed{\rho\dot{\eta} \geq \frac{\rho r}{\theta} - \frac{\nabla \cdot \boldsymbol{q}}{\theta} + \frac{\boldsymbol{q} \cdot \nabla\, \theta}{\theta^2}}. \tag{25}$$